

1. A unit commonly used in astronomy is flux or flux density F_ν . Calculate the flux density in Jy of a small angular size microwave source with an output of 600 W at a distance of 10 m if the power is isotropically radiated and is uniformly emitted over a bandwidth of 10^6 Hz. What will be the flux density if we move the source to the Moon?
2. A “pinhole camera” consists of a small circular hole of diameter d , a distance L from the “film-plane” (see Fig. 1). Show that the flux F_ν at the film plane depends on the brightness field $I_\nu(\theta, \phi)$ by

$$F_\nu = \frac{\pi \cos^4 \theta}{4f^2} I_\nu(\theta, \phi)$$

where the focal ratio is $f = L/d$. This is a crude method for measuring I_ν .

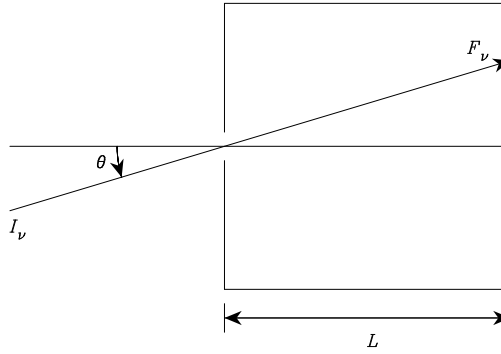


Figure 1: Geometry for a pinhole camera.

3. A certain gas emits thermally at the rate $P(\nu)$ (power per unit volume and frequency range). A spherical cloud of this gas has radius R , temperature T and is a distance d from earth ($d \gg R$).
 - a. Assume that the cloud is optically *thin*. What is the brightness of the cloud as measured on earth? Give your answer as a function of the distance b away from the cloud center, assuming the cloud may be viewed along parallel rays as shown in Fig. 2.
 - b. What is the effective temperature of the cloud? The effective temperature is defined in such a way that $L = 4\pi R^2 \sigma T_{eff}^4$, where L is the luminosity and σ is the Stefan-Boltzmann constant.
 - c. What is the flux F_ν measured at earth coming from the entire cloud?
 - d. How do the measured brightness temperatures compare with the cloud's temperature?
 - e. Answer above questions for an optically thick cloud.

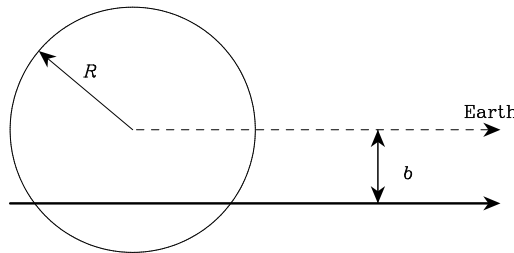


Figure 2: Detection of rays from a spherical emitting cloud of radius R .

4. A spherical, opaque object emits as a blackbody at temperature T_c . Surrounding this central object is a spherical shell of material, thermally emitting at a temperature T_s ($T_s < T_c$). This shell absorbs in a narrow spectral line; that is, its absorption coefficient becomes large at the frequency ν_0 and is negligibly small at other frequencies, such as ν_1 : $\alpha_{\nu_0} \gg \alpha_{\nu_1}$ (see Fig. 3). The object is observed at frequencies ν_0 and ν_1 and along two rays A and B shown in the Figure. Assume that the Planck function does not vary appreciably from ν_0 to ν_1 .

- At which frequency will the observed brightness be larger when observed along ray A ? Along ray B ?
- Answer the preceding questions if $T_s > T_c$.

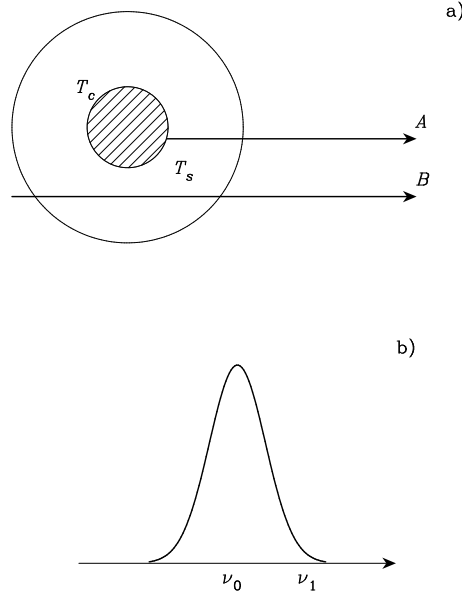


Figure 3: (a) Blackbody emitter at temperature T_c surrounded by an absorbing shell at temperature T_s , viewed along rays A and B . (b) Absorption coefficient of the material in the shell.

5. What is the Rayleigh-Jeans limit? In this limit, write down the radiative transfer equation in terms of brightness temperature. Now you point to a sunlike star and measure its brightness temperature. In between you and the star, there is a cloud in LTE with a uniform temperature of 100 K. The optical depth of this cloud is unknown and can range from 0 to 10. Plot the measured brightness temperature versus the optical depth of the cloud and explain your result.